

Hey, everybody. Welcome to your chapter 14 review. In the next several episodes, we'll take a deeper dive into each of the integrated training components used in the OPT model. We start today with flexibility training concepts. Flexibility training is an important part of a comprehensive fitness program, particularly for clients who participate in tasks that result in performing the same pattern or activity over and over again. This is called pattern overload. And it puts them at risk for accumulative or overuse injury, which can lead to muscle imbalances.

Clients may need flexibility training to correct these muscle imbalances, increase joint Range Of Motion, or ROM for short, and improve the extensibility of muscles and neuromuscular efficiency. The issues that relate to flexibility are surprisingly broad. Muscle imbalance can be caused by postural distortions, repetitive movement, cumulative trauma, emotional duress, poor training technique, poor bodily control, and biased training patterns.

Flexibility also has a major influence on mobility during dynamic motion. Poor flexibility can lead to the development of relative flexibility, which is the process in which the body seeks the path of least resistance during functional movements. It can prevent the nervous system from efficiently recruiting the correct muscle patterns and contractions during movement. It's interesting that, as important as it is, clients don't often come to fitness professionals with the specific goal of improved flexibility.

So it's really important to be able to communicate the benefits of flexibility to your clients and how it relates to achieving their overall fitness goal. To do that, you have to understand the scientific rationale behind flexibility training. At the end of this chapter, you should be able to summarize the importance of flexibility training for clients with differing health and fitness goals, explain the scientific rationale for flexibility training, apply proper methods for effective stretching exercises in client programs, execute, instruct, and cue self-myofascial techniques, static stretching, active stretching, and dynamic stretching exercises using proper and effective methods. Whew. And off we go.

As I said, flexibility training is important to help reduce the risk of muscle imbalances, joint dysfunctions, and overuse injuries. We need flexibility, joint ROM, and neuromuscular efficiency in all three planes of motion-- sagittal, frontal, and transverse. To avoid postural distortion patterns and tissue overload, every segment of the kinetic chain has to be properly aligned. It's a little like if you have a bicycle that's been left out in the rain to rust and one that's been stored carefully, the wheels, chains, and other parts aren't going to work as well on the rusted bike.

The bike that was properly cared for works more efficiently and is at less risk of breaking down. The adaptive potential of the human body is decreased by flexibility limitations and forces the body to move in altered ways, just like a bike with Rusty parts. Poor flexibility can, over time, lead to the development of relative flexibility, which, as described earlier, is the process of our bodies learning to seek the path of least resistance and potentially compensating during functional movements.

It makes sense. The body gets used to being more comfortable within a limited ROM or within altered movement patterns if we don't correct the issue. The Human Movement System, or HMS, also known as the kinetic chain, is comprised of the muscular, skeletal, and nervous systems. The kinetic chain can be classified even further into two regional chains, the upper kinetic and lower kinetic chain.

Muscle imbalances within the kinetic chain can result from altered reciprocal inhibition and osteokinematic and arthrokinematic dysfunction. Now, what does that mean, exactly? To review, osteokinematics describes how the bones and joints are moving through a particular ROM. In other words, it's the joint motions we can see, like flexion, extension, abduction, adduction, or rotation of body segments.

By contrast, arthrokinematics is motion at the joint surfaces, essentially how the bones are moving in relation to each other during a particular movement. There are three major arthrokinematic joint motions-- roll, slide, and glide. Muscle imbalances may lead to synergistic dominance. This is a neuromuscular phenomenon that happens when synergists take over function for a weak or inhibited prime mover or agonist resulting from altered reciprocal inhibition.

This may lead to altered muscle length tension relationships, altered muscle recruitment and force couple relationships, and poor joint surface motion, all resulting in poor movement efficiency. Ultimately, to achieve efficient osteokinematic motion, there has to be good arthrokinematic motion as well as soft tissue flexibility. There is a neuromuscular aspect to flexibility training too.

Neuromuscular efficiency is the ability of the nervous system to recruit the correct muscles, produce force, reduce force, and dynamically stabilize the body structure in all three planes of motion. To allow for optimal neuromuscular efficiency, we have to have proper flexibility in those three planes of motion as well. The scientific rationale for flexibility training is illustrated through the concept of pattern overload and the cumulative injury cycle. And again, this all goes back to movement dysfunctions and muscle imbalances.

But why do we even end up with these dysfunctions and imbalances in the first place? It can be a lot of different reasons or combinations of reasons. They can be caused by poor posture, poor training technique, or a previous injury. Muscle imbalances can also lead to a breakdown in body control over time that can possibly end up in an injury of some kind. You can see how this can become a vicious cycle in the kinetic chain that needs to be addressed.

So here's where flexibility training comes in. It can help with all of these issues and be an excellent preventative measure to keep the human machine running smoothly. The common types of flexibility exercise include self-Myofascial techniques and static, active, and dynamic stretching. Let's take a look at each. We often think of flexibility in terms of stretching, but self-myofascial techniques are also considered when we look at flexibility training. One of the most popular may be Self-Myofascial Rolling, or SMR.

Self-myofascial rolling is thought to produce both a local, mechanical, and neurophysiological effect on the myofascial tissues. This can be done with a foam roller, massage balls, or any number of other cool self-myofascial gadgets out there. These techniques help reduce soreness and even improve joint ROM. The best time to do self-myofascial rolling is before other stretching exercises. However, regardless of your client's level, it's important to be aware of any medical precautions and contraindications that could lead to self-myofascial rolling being a less-than-ideal activity for that particular person.

Moving on to static stretching, this is what we more commonly think of in terms of improving flexibility. Technically speaking, it's the process of passively taking a muscle to the point of tension and holding the stretch for a minimum of 30 seconds. By holding the muscle in a stretched position for a prolonged period, the muscle spindle becomes inhibited, resulting in a relaxation response. This provides for a better lengthening reaction.

In fact, static stretching may go even further than you think and impact many sensory mechanisms within the nervous system to facilitate a greater stretch tolerance. This form of stretching also works to increase joint ROM. Static stretching should be used to decrease the muscle spindle activity of an overactive muscle before and after activity and to stretch muscles that have been identified as overactive during your assessment process.

Next, active stretching is the process of using agonists and synergists to dynamically move the joint into a ROM, typically holding for one to two seconds and repeating for 5 to 10 repetitions. This form of stretching increases motor neuron excitability, creating reciprocal inhibition of the muscle being stretched. An example of this process would be contracting the outer hip gluteal muscles to move into a side lunge position to stretch the adductor complex of the same leg.

By contracting the gluteals as the leg is moved into abduction during the lunge, the adductor muscles are getting reciprocally inhibited, which causes them to relax, making it easier for them to lengthen mechanically during the stretch. Active, isolated stretches are suggested for pre-activity warm-up, like before a sports competition or high-intensity exercise.

If a client has muscle imbalances, active stretching should be done progressively after self-myofascial release and static stretching for muscles determined as overactive during the assessment process. Finally, dynamic stretching uses the force production of a muscle or group of muscles and the body's momentum to take a joint or multiple joints through the full available ROM.

Dynamic stretching employs the concept of reciprocal inhibition to improve soft tissue extensibility. Clients can perform one set of 10 repetitions using anywhere from 3 to 10 different dynamic stretches. Dynamic stretching is also suggested as a warm-up before athletic activity. If your client does show some muscle imbalances, self-myofascial techniques and static stretching should precede dynamic stretching for any overactive muscles you've identified.

It's also a good idea to ensure the client has good levels of tissue extensibility, core stability, and balance capabilities before going for an aggressive dynamic stretching program. With dynamic stretching, it's important to adequately prepare the client for activity while also avoiding fatigue. The goal is to prepare the body for more intense activity rather than cause their energy and endurance to crash before they even get to the primary workout.

I also want to mention that although flexibility training has tons of benefits, there are certain types of stretches that are considered controversial and could increase the risk of injury. Most stretches are safe when performed correctly with proper posture and technique. But it's important for you to have a good understanding of flexibility methods and types of stretching that could be less safe due to the body's positions that might be required.

You should also be familiar with any specific contraindications associated with, let's say, a client's current condition or past medical history that might preclude them from specific flexibility exercises and moving through specific ranges of motion. For example, if you have a client who recently had a hip replacement, there are certain positions and stretches that should be avoided so as not to disrupt their healing and recovery.

So be mindful of the importance of flexibility training. Take advantage of the information and specific flexibility building exercises that NASM has offered as you build programs for your clientele. More to come on other training concepts-- be well, and happy studying.